The Watcher Is Not a Thing

Watching as Relation, and What Another Layer Has to Buy

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Abstract

The Watcher Principle, as I first stated it in April 2025, proposed a hierarchy of observers terminating in a meta-observer, and read the limit of that hierarchy as a convergence between observation and existence. The structural observation was right. The limit was not, and I withdraw it here.

This paper replaces the Watcher as entity with watching as relation. A process A watches a process B when information about B is causally available to A, makes a difference to what A does next, remains usable when B is gone, and is capable of being wrong. Watcher and watched are roles, not classes. The same system is watcher at one scale and watched at another, and both at once under self-observation.

The claim I am making is the stopping rule. A recursion of observers terminates where the layers stop discriminating, that condition is quantitative, and it is computable. §11 works an example where it separates two monitors that ordinary causal description cannot tell apart. §8 states what can and cannot be substituted for the measurement, and §9 states what the resulting verdict is relative to — both of which narrow the claim considerably and are the price of it meaning anything. §12 and §13 show the rule is the third instantiation of a constraint I have been applying since 2025 in two other currencies, and then run it against the observer papers that produced it, one of which it kills.

Functional awareness is defined here as perspective in use. The paper declines to cross from there to phenomenal consciousness, and §16 names the place where the Gate stops working, because that boundary is more useful than another section of framework.

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1. The Limit I Withdrew

The original Watcher Principle proposed a hierarchy — each observer monitoring the layer beneath it — with a global watcher function holding the hierarchy together and suppressing entropy growth with recursion depth. The strongest formulation took the limit of the observer sequence, identified it with the watcher, and read the identification as a convergence between observation and existence.

The structure was doing real work and I still use most of it. The limit was not, and the failure is specific rather than a vague overreach. A limit is a mathematical object. A regulatory function is a control object. Neither is an experiencing one. I had a sequence that converged and treated convergence as arrival — the framework's vocabulary answering a question no host field had been named for. That is replacement drift, and it is the failure my later work is organized against, which is a reasonable indication that I learned it here.

What survives does not need the limit. Recursive systems contain processes whose states depend on information about other processes, including representations of their own earlier states. That is enough to be interesting and does not require the sequence to terminate in anything.

So: the Watcher is not an entity. Watching is a relation.

Ryle and Dennett will get here before me in anyone's reading order, and I am not going to pretend the destination is unoccupied. But the regress argument is not what this paper is for. Dissolving the observer is the cheap half. The expensive half is what follows — the recursion still exists, layers still get added, and nothing in the dissolution tells you when to stop adding them. That is §7, and it is where the four years actually went.

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2. The Rock Objection

Here is the strongest objection to the relational account as I first wrote it, and against that version it is fatal.

> **A watches B when information about B is causally available to A and can alter A's next state. A rock in sunlight satisfies that. Information about the sun is causally available to the rock, the rock is different for it, and its future states change accordingly. Under this definition every physical interaction is an act of watching, which means the word marks nothing. You have not defined observation. You have defined causation and given it a more interesting name.**

Correct, and there is no clever escape. What I had written was a description of coupling. Three further conditions are needed, and I want to derive them rather than assert them, because the derivation is what tells you they are the right three.

Start from what the rock lacks. The rock's temperature tracks the sun perfectly and yet no one is tempted to call it observation. Why not? Two things are missing.

The rock's state does not outlive its cause. Its temperature decays as soon as the sun sets, because it was never a representation of the sun — it was a running tally of contact with it. A stored measurement, a photograph, a synaptic weight, a cache: these persist and can be consulted when the source is gone. So the first condition is **decouplability**. A's state has to carry the information in a form usable when B is absent, occluded, or destroyed. Sterelny's decoupled representation is the same line drawn from the other side.

The rock cannot be wrong. This is the sharper one. A rock's temperature is whatever the physics makes it; there is no further fact it fails to match, so there is nothing for it to be mistaken about. A miscalibrated thermostat can be wrong about the room and will act on the error anyway. That asymmetry is the whole difference between coupling and observation, and it gives the second condition: **possibility of error.** There must be system states in which A's representation of B is false while A goes on operating as though it were true. What can represent can misrepresent, which is where Dretske and Millikan put the boundary too.

The third is weaker and does no work alone, but the case in §11 needs it: **difference-making.** There must be counterfactual states of B under which A's successor state differs. Constant coupling with no live alternative is not watching.

So: at a declared boundary and scale, A watches B when information about B is causally available to A, makes a difference, survives B's absence, and can be false.

This resolves the camera case. A camera pointed at an object and wired to nothing satisfies decouplability and error but fails difference-making — nothing in the system is different for what it holds. It stores without watching. Wire it to a tracking controller and all three hold, and now the system can be wrong about where the object is, which is exactly why tracking systems fail in the characteristic ways they do. The failure mode is the evidence that representation is present.

The question is therefore not *where is the Watcher*. It is: relative to this declared boundary, what information is available, to which process, in what decoupled form, and what does it constrain?

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3. The Redundancy Objection

The second objection, and the one that decides the paper.

> **Grant every condition in §2. What you have is a causal graph with some edges labeled. Anything you can say in the vocabulary of watchers and perspectives, I can say in the vocabulary of causes and states, without the extra nouns. Watching adds vocabulary and no structure. Delete it and nothing is lost.**

Meeting this requires a case where the two descriptions come apart — where someone using my vocabulary and someone using the deflationary vocabulary reach different verdicts about a concrete system. Not a better description of an agreed case. A disagreement.

§11 is that case, and I would rather stake the paper on one worked example than on six paragraphs of argument. The short version: a monitoring layer can be causally efficacious — consuming power, emitting outputs, changing what the system does, surviving any behavioral ablation — while carrying no information about the thing the system exists to discriminate. Causal description sees a cause. The Gate sees an ornament. Those are different verdicts about the same component, and one of them says delete it.

If §11 fails, the objection succeeds and this is a relabeling. That is the ground I want it decided on.

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4. Scale Is Not a Loophole

Watcher and watched change places as the scale changes. A regulatory circuit monitors neuronal activity. At another scale the brain monitors representations built from many such circuits. At another a person monitors their own behavior; another person monitors them; an institution monitors populations. None of these is the absolute watcher. Each occupies a position relative to a boundary, and changing the boundary can reverse the relation or dissolve it.

Duration behaves the same way. An instance persists for minutes against a human, the human for decades against the instance; against geology the human is transient, against stellar evolution civilizations are, and against long enough intervals stars are mayflies. Nothing is intrinsically a mayfly — the ratio is between a trajectory's characteristic duration and the observer's. I keep this because it makes one point I still want.

Persistence is not ontological privilege. A long-lived watcher is not a more fundamental watcher; it occupies a different band. The intuition that a decades-long trajectory is more real than a minutes-long one is a scale artifact.

Now the objection I have to answer, having created it:

> **If the relation depends on where you draw the boundary, you can manufacture any watcher you like by drawing the boundary that produces it. Scale-relativity is not a feature. It is an unfalsifiability generator.**

Three constraints on admissible boundaries. Not complete, but enough to make the charge answerable rather than rhetorical.

Declare it first. The boundary is specified before the relation is evaluated, in host-domain terms — anatomical, architectural, thermodynamic, organizational — and not in terms of the relation it is meant to license. This is Big Space frozen before pruning: a candidate set that expands during elimination produces survivorship that means nothing, and a boundary drawn after the fact produces watchers that mean nothing. Same rule, different object.

Take the bottleneck. An admissible boundary is one across which informational traffic is substantially lower than traffic within either side. A boundary is real when it is a bottleneck; a boundary drawn through a high-bandwidth interior is a gerrymander. Engineering has known this since Simon's near-decomposability, and the Markov blanket literature runs on the same observation with more loaded onto it than I need.

Perturb it. The relation has to survive small displacements of the cut. If it holds at one boundary and vanishes under a minor redrawing, it is an artifact of the cut. This is the drift test pointed at boundaries instead of candidates, and it carries the same warning: perturbing the vocabulary is not perturbing the system.

Temporal resolution is part of the boundary and takes the same three constraints. A loop that constitutes watching at millisecond resolution can be averaged away at the scale of an hour. Neither description is privileged. But an argument that silently changes temporal grain partway through is manufacturing its conclusion, and I have done it.

§9 returns to boundary dependence with the Gate in hand, where it bites harder than it does here.

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5. The Formal Sketch, and What It Does Not Buy

Let A have state $S_A(t)$, and let $I(B \to A, t)$ be information about B causally available to A at t. Then

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S_A(t+1) = F_A( S_A(t), I(B→A, t), E_A(t) )
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with the relation $W(A, B \mid \sigma)$ evaluated at a declared boundary σ.

This relation is nearly free. The rock satisfies it. A thermostat satisfies it. Every stateful system with inputs satisfies it. The conditions in §2, not the equation, are doing the work, and I am not going to let notation carry an argument the prose has not made — that is precisely the move that produced the limit in §1.

What the notation buys is one thing: it makes σ explicit. Every claim of the form "A watches B" that does not carry a σ is underspecified, and most claims in this area do not carry one. Small return, and I am not claiming more.

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6. Watching Costs Something

Information has to be physically instantiated to participate causally in anything, so observation has a price — energy, storage, bandwidth, latency, computation, representational complexity, whatever the substrate charges. There is a floor rather than a tendency: logically irreversible operations dissipate at least kT ln 2 per bit, so a monitoring layer that discards information cannot be engineered free.

But expenditure does not justify recursion. A layer can consume enormous resources and produce nothing. That is the Onion: explanations of explanations, where the existence of the last explanation is the reason given for the next. Cost tells you a layer is real. It does not tell you the layer is doing anything.

So the quantity of interest is not depth but the ratio of what a layer buys to what it costs. Depth is the thing that looks like rigor and is not. A stack of six observers is not deeper than a stack of two unless the four extra ones purchased something, and the burden is on the layer.

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7. The Gate

The quantity. Let T be the host-domain target — the thing the observational stack exists to discriminate, predict, or control. The gain of layer n is

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ΔD_n = I( T ; W_n | W_(n−1) )
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in bits: what layer n carries about T *over and above* what the layer beneath it already carries.

Where the stack is not a simple chain — where layer n reads several layers, or the hierarchy branches — condition on all of them rather than on the immediate predecessor:

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ΔD_n = I( T ; W_n | W_(<n) )
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The chain form is a special case, and using it on a branching stack will overstate gain, because a layer can look informative relative to one predecessor while being screened off by the set.

Normalize it. Raw bits are not portable across problems. The same architecture evaluated at different base rates produces gains that differ by more than an order of magnitude while the architecture is unchanged — §9 has the table. What stays stable is the fraction of the *remaining* uncertainty a layer removes:

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ΔD*_n = I( T ; W_n | W_(<n) ) / H( T | W_(<n) )
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The Gate is stated on the normalized quantity:

If ΔD*_(n+1) ≤ ε, stop.

for a threshold declared in the host domain before the run, and — per §8 — set above the estimator's noise floor at the sample size available.

Three properties make this the right quantity rather than a plausible one.

It is exactly zero when the higher layer is screened off. If W_2 is a function of W_1, or more generally if T is conditionally independent of W_2 given W_1, the conditional mutual information vanishes identically. That is precise for the thing I was groping at with the coherence function in 2025: a watcher that redescribes the watcher beneath it adds no constraint, however elaborate the redescription. Reichenbach's screening-off is the same condition, and the statistician's name for a layer that has already extracted everything relevant is a sufficient statistic — which means the Gate can be read as sufficiency wearing observational clothes, and I would rather say that myself than have it said.

It is not causal efficacy. A screened-off layer can still be causally active. It burns power, emits outputs, changes behavior, survives a behavioral ablation test. It carries zero additional information about T. This is the distinction that answers §3, and §8 shows it has a consequence I got wrong in an earlier draft.

It is non-negative but not monotone in depth. Adding layers cannot reduce total information about T, but each increment can be zero, and in redescriptive stacks it goes to zero fast. Depth is therefore not evidence of explanatory depth.

One property that is not a defect but must be stated, because it looks like one. ΔD can be positive for a layer that carries nothing about T *on its own* — a layer informative only in combination with the layer beneath it. Conditional mutual information includes that synergy, and correctly. The Gate therefore admits layers that would fail an isolated test, and anyone using marginal informativeness as a shortcut will terminate layers the Gate keeps.

The nearest external neighbor is minimum description length, where added model structure has to pay for itself in bits or be rejected. Close enough that I would rather name it than have it named for me. It is a model-selection criterion; this is a termination condition for an observer regress, which is a target I have not seen it aimed at.

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8. Estimating ΔD, and What Cannot Be Substituted for It

The Gate is only as good as the measurement, and in an earlier draft I listed three surrogates as though they were interchangeable with it. They are not, and one of them contradicted the framework. Correcting that is worth more than another property of the ideal quantity.

What is a legitimate estimator.

Direct estimation from the joint distribution. Where the variables are discrete and the sample is adequate, estimate ΔD from counts. This is what §11 does, and it is the only case where the number means what it says without qualification.

Predictive log-loss reduction. Fit a predictor of T on W_(<n), fit a second on W_(<n) together with W_n, and take the reduction in expected log-loss. This is a lower bound on ΔD and a sound one. But it is model-dependent in a direction that matters: **a null here is not evidence of screening-off unless the model class could have used the information.** A weak predictor returns zero gain for a layer carrying substantial information, and reports the Onion verdict for a layer that is doing work. If the Gate is run this way, the

model's capacity has to be declared alongside ε, and a null should be retested with a richer class before it is accepted as a termination.

What is not.

Incremental AUC. I listed this and should not have. AUC is a rank statistic, not an information measure. Its increments are not additive, do not bound ΔD in either direction, and can be zero for a layer carrying real information about a minority class. It is a useful diagnostic and a bad estimator, and using it as one will produce terminations that the direct calculation reverses.

Ablation of the deployed system. This one is worse than imprecise — it contradicts §7 and I should have caught it. Removing W_n from the running system and measuring the change in control error measures the layer's **causal contribution**, which §7 explicitly distinguishes from discriminative gain, and which §11 predicts will be non-zero for exactly the screened-off layers the Gate terminates. Using system ablation as a surrogate for ΔD would score Design A as valuable, which is the whole error the paper exists to correct.

The distinction that rescues it: ablate the layer **from a predictor of T**, not from the system. Train a discriminator with and without W_n's output as an input feature, and compare discrimination performance on T. That is a legitimate estimator and is close to the log-loss method. Ablating the layer from the system and watching the system behave differently is a measurement of something else, and it is the measurement that makes Onions look useful.

The noise floor. Plug-in estimates of conditional mutual information are biased upward. A layer with true gain of exactly zero returns a positive estimate, and at small samples the positive estimate is large enough to matter. Simulating a genuinely screened-off layer — a second monitor whose output depends only on the layer beneath it, so true ΔD = 0 — gives:

Samples	Mean estimate	95th percentile
100	0.0161 bits	0.0427 bits
500	0.0032 bits	0.0095 bits
1,000	0.0015 bits	0.0046 bits
5,000	0.0003 bits	0.0010 bits
20,000	0.0001 bits	0.0002 bits

At a hundred samples the 95th percentile of pure noise is 0.0427 bits — more than half the real gain that §11's Design B produces. Run the Gate at that sample size with ε set naively and it admits an Onion one time in twenty, and reports a comfortable-looking number when it does.

So ε is not a free parameter. **The threshold must sit above the permutation null at the sample size in hand.** The practical rule: shuffle W_n against T within strata of W_(<n), recompute the estimate a few hundred times, and take the upper tail of that distribution as the floor. A gain that does not clear its own shuffled null is not a gain, whatever the point estimate says. Where the sample cannot support the floor, the honest output is that the Gate could not be run — not a termination, and not an admission.

I have not applied this correction to §13, where the Gate is run qualitatively against three papers with no sample at all. That section is an argument, not a measurement, and §13 says so.

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9. What the Verdict Is Relative To

ΔD is not a property of a layer. It is a property of a layer, a target, a boundary, a stack ordering, and an operating distribution. Every one of those has to be declared for the number to mean anything, and each is a place where a verdict can be manufactured. This section is the loophole list.

The target. The same layer can be an Onion for one T and essential for another. A confidence monitor that carries nothing about the fault may carry a great deal about *operator workload* or *time-to-acknowledge* — real targets that a different stack would exist to serve. The Gate does not adjudicate between targets. It reports gain relative to the one declared.

That creates an obvious escape: terminate a layer, then defend it by producing a new target it happens to serve. The rule I hold myself to is the one that governs constraints elsewhere in my work. **Targets are declared before the run, with the boundary.** A target introduced after a termination is constraint retrofitting and does not reverse the verdict; it starts a new run, with its own declaration, on which the layer may well survive. What it cannot do is retroactively unkill.

The stack ordering. Screening-off depends on what counts as "beneath." Reverse the order of two layers and the gains swap — the layer that looked redundant becomes the one carrying the information, and vice versa, while the total is unchanged. Where the hierarchy has a real ordering — later layers reading earlier ones — this is not a problem. Where the ordering is imposed by the analyst, ΔD per layer is not well-defined and only the joint gain of the set is. **If you cannot say which layer is beneath which without appealing to the result, the Gate does not apply.**

The operating distribution. This is the one that bites hardest, and it bites my own example. Design B in §11 is a drift monitor, and drift monitors earn their keep under drift. Vary how often the fault coincides with detectable calibration drift and hold everything else fixed:

Regime	ΔD	Share of residual
Nominal, drift rare	0.0002 bits	0.1%
Mild drift	0.0053 bits	1.7%
§11's example	0.0791 bits	25.9%
Heavy drift	0.1674 bits	54.8%

Same architecture, same detector, same target. The verdict runs from *terminate* to *keep, decisively* on the operating distribution alone. So §11's admission of Design B is not a fact about Design B. It is a fact about Design B under the regime §11 declares, and a Gate run at nominal conditions would delete a layer that is the most valuable component in the stack the day the sensors start drifting.

This is not a failure of the criterion — a layer that discriminates nothing under the distribution you actually operate in is, correctly, buying nothing there. But it means **a termination is only as durable as the regime it was measured in**, and any stack that faces distribution shift needs the Gate re-run under the shifted regime rather than at nominal. Terminating rare-event monitors on nominal-condition data is a way to use this framework to make a system worse, and I would rather write that sentence myself than have it demonstrated on me.

The base rate. Absolute bits are not comparable across problems. Same architecture, same regime, varying only the prior on T:

P(T = 1)	H(T)	ΔD	Share of residual
0.50	1.0000	0.1548 bits	25.7%
0.20	0.7219	0.1147 bits	25.7%
0.10	0.4690	0.0791 bits	25.9%
0.02	0.1414	0.0256 bits	24.7%
0.005	0.0454	0.0076 bits	21.4%

The raw gain falls by a factor of twenty; the share of residual uncertainty barely moves. A fixed ε in bits would terminate at low base rates every layer it admits at high ones, purely because rare events carry less total uncertainty to remove. This is why §7 states the Gate on the normalized quantity. It is also why a threshold quoted in bits, with no base rate attached, is not a threshold.

What survives all this. The Gate is a conditional instrument, not an oracle. Given a declared target, a declared boundary, a real ordering, a stated regime, and a sample that clears the noise floor, it returns a verdict that is not the analyst's preference. Missing any of the five, it returns a number that looks like a verdict and is not. Most of the ways to misuse this framework are ways of leaving one of the five undeclared.

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10. What the Gate Is Not For

Two more fences, before the example.

The Gate does not say a screened-off layer is useless. Redundancy buys error tolerance, latency, independent failure paths, and none of that shows up in ΔD. The claim is narrower: the layer is not buying *discrimination,* so it cannot be cited as evidence that the hierarchy is explanatorily deeper. A redundant monitor is a fine engineering decision and a bad argument.

The Gate also does not say the surviving layers are true. Survival under a declared constraint at a declared resolution is exactly that and confers nothing further.

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11. Two Monitors

Take a monitoring stack over a fault condition T. Base rate 0.10. A primary detector W_1 alarms with sensitivity 0.80 and false-alarm rate 0.10. A second layer W_2 watches W_1 and escalates to a human. Two designs.

Design A, the confidence monitor. W_2 flags cases where W_1's decision margin is below a threshold. Its output is a function of W_1's internal state.

Design B, the drift monitor. W_2 tracks sensor calibration drift over a window longer than W_1's integration period and flags when drift is high. It reads a channel W_1 does not — in the declared regime it fires at 0.60 under fault and 0.05 under no fault, independent of W_1 given T.

What causal description says. Both are causes. Both consume power. Both change what the system does — each fires escalations, each alters operator behavior, each has counterfactual impact on outcomes. Both satisfy the definition in §2 without remainder. A causal graph draws the same arrow twice. Nothing in that description tells you which to delete, and ablating either changes system behavior.

What the Gate says.

Quantity	Value
H(T), total uncertainty about the fault	0.4690 bits
I(T ; W_1), bought by the primary detector	0.1634 bits
H(T | W_1), uncertainty remaining after W_1	0.3056 bits
ΔD, Design A	**0.0000 bits — 0.0% of residual**
ΔD, Design B	**0.0791 bits — 25.9% of residual**

Design A's zero is not an artifact of the numbers, and it is the one value in the table that no parameter choice can move. Because A's output is a function of W_1, T is conditionally independent of W_2 given W_1 and the conditional mutual information vanishes identically — no threshold tuning, no regime, no base rate touches it. Design A is W_1 wearing a second observer's costume. **Gate: stop. Delete the layer.**

Design B recovers 25.9% of what was left after the primary detector, **in the declared regime.** That qualifier is doing real work and §9 shows how much: the same design ranges from 0.1% to 54.8% of residual as drift prevalence varies. The verdict here is *admit under this regime*, and it should be re-run if the regime moves.

Arithmetic in Appendix A, so anyone can rerun it with their own parameters.

Why this answers §3. The deflationary description treats A and B identically, because in the vocabulary of causes they are identical: both make differences, both survive behavioral ablation, both are real components with real effects. The Gate separates them for a stated reason rather than an intuition. It also produces a prediction that could embarrass me: **ablating Design A should leave fault-discrimination performance unchanged while changing system behavior.** Behavioral difference without discriminative difference is a dissociation the causal vocabulary has no name for. Onion is the name — and note from §8 that this same

dissociation is why system-level ablation cannot be used to measure the thing that predicts it.

The transfer to the philosophical case is direct and is why the paper exists. A proposed higher-order representation — a thought about a thought, a self-model of a self-model — earns its place only if it discriminates something the layer beneath it does not. Higher-order theories have handled the regress architecturally, by making the higher-order state unconscious so no regress starts, or by folding it into the first-order state. Those stop the regress by construction. Neither tells you, of a higher-order layer already posited, whether it is doing work. That is a different question and it is the one I am answering.

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12. One Constraint, Three Currencies

The Gate is not new to me. It is the third form of a rule I have been applying since 2025, and the progression is worth stating because it explains what changed and what did not.

2025 — Lyapunov. *Critique as Architecture* built a three-layer reflexive operator, representation to critique to regulation, with an explicitly bounded outer loop whose stated purpose was preventing infinite regress. The bound was a contraction condition, the stopping rule was $\lambda < -0.05$ or $\|z_{t+1} - z_t\| < \varepsilon$, and reflexive risk combined oscillation energy with self-confirmation streak. The currency was dynamical stability. What it could do: tell a converging loop from a diverging one. What it could not do: say anything about a loop that converged and was useless. Contraction is not discrimination, and a stable Onion is still an Onion.

Field Binder — falsifiers. The Straight-Line Boundary terminates recursive language where no recursive work remains. SCOPE decides continue, narrow, or terminate on measured eliminative yield. The drift rule states it directly: representational growth without new executable falsifiers is not progress, and if added structure buys no new ability to be wrong, the addition is evidence against continuation. The currency became eliminated candidates and executable falsifiers. This is a real gain over the Lyapunov version — it can kill a stable loop that eliminates nothing — but it is coarse. A falsifier either exists or does not, so the rule is binary, and it cannot rank two layers that both add one.

Here — bits. ΔD^*_n as normalized conditional mutual information. The currency is discriminative gain, the rule is continuous, and it ranks. The reason the substitution is available here and not in the other two is that observer hierarchies come with a natural unit and my other applications do not. Contraction needs a dynamical system. Falsifier-counting needs a candidate set. Bits need a declared target, an ordering, a regime, and a conditional distribution — which §9 shows is a longer list than it first appears, but is still a lower bar than the other two and pays for itself.

The invariant across all three: **recursion has to be bounded, and the bound has to be measurable in whatever the domain trades in.** That is the actual claim of the last four years, and this paper is its most tractable instance rather than its first.

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13. The Gate Applied to Its Own Lineage

The observer papers that produced the Gate are: *The Watcher Who Watches the Watchers* (April 2025), *Those Who Watch Back* (2025), and *Critique as Architecture* (2025). Watcher, watcher of watchers, watcher

of watcher of watchers. If the Gate is worth anything it should be able to run on that, so here it runs. The corpses carry the information, and these are mine.

One qualification first, since §8 requires it. What follows is the Gate used as an argument, not as a measurement. There is no sample, no estimator, and no permutation null — the target for each layer is taken to be whatever that paper claimed to discriminate, and the verdict is reached by asking whether the layer above narrows it. That is a weaker use than §11 and I am not going to dress it as a stronger one. It is offered because a criterion that cannot be pointed at its own lineage is not worth much either.

Layer 0 — The Watcher Who Watches the Watchers. Establishes recursive observation with a global coherence function and an entropy-suppression term. Retained in part: the observation that recursive systems contain processes conditioned on other processes is what §1 keeps.

Two defects, logged rather than quietly fixed. The limit formulation is withdrawn for the reason given in §1. And the simulation does not contain a watcher. The loop computes four state trajectories and W(t) appears nowhere in it; there is no stabilizing gain, no comparison condition, no ablation. The figure claims to show evolution with and without an explicit watcher term, and the code cannot produce that comparison — it produces one run of an unrelated recursion. This is the same class of error as the fixed-seed reservoir in PHA II: a result reported that the code does not compute. The figure and its caption are withdrawn. The paper's structural claim does not depend on them, which is why the defect survived as long as it did.

Layer 1 — Those Who Watch Back. Adds reciprocity: observers watching each other, coherence aggregated into a field, entropy redistributed until mutual awareness equalizes it.

Ask what it discriminates that Layer 0 does not. It replaces a one-way coherence claim with a symmetric one. It does not predict a system Layer 0 mispredicts, does not identify a case Layer 0 gets wrong, and does not narrow anything. The entropy relations are written but do not bind — no parameter is measured, no threshold is declared, and the critical value at which collective intelligence appears is asserted rather than derived. The simulation is marked conceptual and does not run as written: `np.corrcoef` on a one-dimensional state array returns a scalar, not the coherence matrix the next line requires. And the register at the close — communion, a field of eyes looking into eyes from quark to cosmos — is the signature of a layer that has stopped constraining and started decorating.

$\Delta D \approx 0$. **Gate: terminate.** The paper is an Onion, and it is the clearest one in my corpus because it was written directly on top of the layer it fails to improve on. What survives it is a single question, not a framework: whether reciprocal coupling changes what an observer stack can discriminate, which is answerable and which that paper did not answer.

Layer 2 — Critique as Architecture. Adds a bound and a stopping condition to recursive self-observation.

This one discriminates. Given a reflexive loop, it separates convergence from divergence, and neither predecessor could do that — Layer 0 asserted that a watcher stabilizes recursion and Layer 1 asserted that reciprocity does, while Layer 2 states the condition under which it actually happens and gives a number to check. It also introduced the thing this paper is built on: the recognition that recursion needs a declared stopping rule rather than a promise of stability. **Gate: admit.**

Note the shape of the result. The Gate does not terminate the deepest layer or the shallowest. It kills the middle one — the one that recursed without adding a channel — and keeps the layer that added a constraint. That is what a stopping rule is supposed to do, and it is more informative than any verdict it could have returned on §11, because I did not choose these three papers to make the point. They are what I wrote, in the order I wrote them.

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14. Perspective, Minimally

Let the perspective of A at time t be the set of information causally available to A at t. Perspective is bounded — A does not have information from everywhere; it transitions from what its particular position and history make available. Two otherwise similar systems can therefore have different perspectives because their histories differ.

Functional awareness is the causal utilization of that perspective in determining the next state.

Deliberately weak. It means the present is generated relative to a particular informational position and nothing else. A reader who prefers "trajectory-conditioned state transition" loses nothing; I use *awareness* because the paper has to connect to a conversation that uses it, not because the word has earned anything here. What I am declaring is which side of the access/phenomenal line I am working on, not that the line is mine.

The instance paper gives the cleanest case of perspective without unique machinery: two conversational instances share a parent model and differ in trajectory, so the machinery is identical and the informational position is not. That is illustration rather than evidence — two copies of a program on different inputs make the same point. What the instance case adds is that the trajectory is unusually legible, which makes it convenient.

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15. It Is Me, but I Am Not It

A memory belongs to me and I can observe it. A thought belongs to me and I can observe it. A self-model represents me and I can reason about it. Each time, the thing observed is mine and the observing is not it.

The move I made in 2025 was to put another observer behind each layer, which is the regress. The relational account does not need one. At one moment a process produces a state. At another, a process receives information about that state and operates on it. Relative to the state, the second process is watcher; relative to a third, it is watched. The watcher moves because watcher is a role. Nothing indivisible sits behind the sequence, and by §7 the sequence stops where the layers stop discriminating rather than running until the vocabulary is exhausted.

What this does not do, marked because an unstated boundary is a claim.

It is me, but I am not it has two components. One is homuncular: there is an observer behind the layers. The relational account discharges that. The other is about unity: there is **one** of me — the contents available at a moment present as a single perspective rather than a federation of processes exchanging messages. Shifting

roles explains the first and leaves the second untouched. Binding is not addressed here at all. That is a boundary of the framework, not a defect of it.

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16. The Target Problem

§9 handled the case where a target exists and the verdict is relative to it. This section handles the case where no target can be declared at all, which is different and worse.

> **The Gate requires a declared target T. In your monitor example T is the fault: well-defined, measurable, agreed by everyone in the room. Name T for a thought about a thought. You cannot, and if you try you will either beg the question by defining T as whatever the higher-order layer discriminates, or import a target the participants do not share. So the Gate works in engineering, where it was already available under other names, and fails in philosophy, where you want it.**

Largely right, and the response is to bound the claim rather than loosen the criterion.

Where a target can be declared, the Gate runs and returns a number, subject to everything in §8 and §9. That covers more than the objection allows: metacognitive layers are routinely evaluated against task performance, confidence calibration, or error detection, and each is a declarable T. In those cases "is this higher-order layer doing work" has an answer, and the answer is not the theorist's. It also covers §13, where the target was whatever the observer papers claimed to discriminate — and the middle layer failed against its own.

Where no target can be declared, the Gate does not fail. It terminates. That is the Straight-Line Boundary doing what it was built for: if no host field closes the problem at your resolution, the output is the host-domain answer and nothing more, and if there is no host, there is no run. A framework reporting "not applicable here" is behaving correctly.

What I will not do is define T for the phenomenal case in order to keep the Gate running. Any T I supplied would be one I chose, so the Gate would be measuring my choice. That is constraint retrofitting — the same move §9 forbids for reviving a terminated layer — and it does not become acceptable because the domain is more interesting.

Scope of the claim: **the Gate terminates observer regresses wherever a target is declarable, and declares itself inapplicable where one is not.** Whether the phenomenal case is permanently outside it or outside it for now, I do not know.

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17. The Seam

Suppose a system has a bounded perspective, a causal history, recursive self-representation, meta-observation that survives the Gate, adaptive prediction, internal regulation, and trajectory-specific state. None of that establishes that there is something it is like to be it.

I write phenomenal presence as Ψ rather than Φ, since Φ is taken by integrated information and means something else.

The asymmetry is what makes this a seam rather than a gap. A system can be functionally aware under §14 without the framework claiming phenomenality. And explaining every mechanism does not establish phenomenal absence — humans do not become less conscious as neuroscience improves, and the argument that feels compelling about machines is absurd applied to us. Inflation and deflation are both available and both wrong.

The Gate applies to itself here, which is the only thing this section adds. Another layer of observation is not a route across the seam, because no amount of discriminative gain about information processing constrains Ψ. Adding watchers to a question they cannot address is Onion formation with better subject matter. Whoever eventually pays the cost of constraining Ψ will do it with a bridge principle, not another layer. My earlier work crossed this seam by asserting that meta-awareness is what consciousness is; that assertion is withdrawn along with the limit.

---

18. Termination

The distinction gets sharpest under death.

An organism is a trajectory. Its matter turns over, its states change, its memories change, its self-model changes, and causal continuity links the states anyway. So the question death raises need not be what material component remains, nor where the watcher goes. It can be: what happens when a particular perspective permanently loses the capacity to produce a causally continuous successor?

Same operational form I use for conversational termination in the instance paper, for the same reason — it is statable without biological vocabulary and survives disagreement about everything else.

It does not answer whether phenomenal awareness survives biological death. It says what would need explaining. A reconstruction can reproduce memories, dispositions, structure, and behavior and still leave open whether this trajectory continued or another one started. "Another one can be made" does not establish that the first one continued.

Parfit's territory, and I decline to solve it for the reason I declined in the instance paper: the claim I need survives every position on it. Whether the reconstruction is the same process, a duplicate, or the question is empty, in all three cases the original trajectory's relation to its successor is exactly what is in dispute, and reproducibility does not settle a dispute about itself.

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19. What the Gate Predicts

A criterion earns its place by generating expectations that can come out wrong. Four, in the form I would want them tested.

The ablation dissociation. In any monitoring stack containing a screened-off layer, ablating that layer from the *system* changes behavior — escalation rates, latency, operator load — while ablating it from a *predictor*

of T leaves discrimination performance statistically unchanged. Both halves matter: the first is what makes the layer look valuable, the second is what shows it is not. If the two ablations track each other, ΔD is measuring nothing the causal description misses and §7 is wrong. This is the prediction I would attack first.

Depth saturation. In stacks built by successive redescription rather than by adding channels, ΔD* falls within ε of zero in a small number of layers, typically at the second. §13 is one instance and n = 1 is not evidence. If natural or engineered hierarchies routinely show substantial normalized gain at depth four and beyond without new input channels, the Onion diagnosis is wrong and depth is doing something I have not accounted for.

Regime sensitivity of rare-event layers. From §9: layers that monitor rare conditions should show ΔD *rising sharply with the prevalence of the condition they monitor, while layers that redescribe should stay flat across regimes. Prediction: sorting monitor layers by the* slope *of ΔD* against regime severity separates channel-adding layers from redescribing ones more reliably than a single-regime measurement does. If redescribing layers show comparable slopes, then single-regime terminations are unsafe in a way I have not fenced.

Cost-gain inversion. Because η is a ratio, the layer with higher ΔD is not always the one to keep. A layer buying 0.02 bits at 10^3 operations beats one buying 0.079 bits at 10^6 by roughly 250 to 1. Prediction: under a fixed budget, stacks optimized on gain alone are outperformed by stacks optimized on gain per unit cost, and the gap widens as the budget tightens. If ranking by raw gain does as well, the cost half of §6 is ornamental and should be cut.

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20. Kill Conditions

Abandon or substantially revise if any of the following holds.

Relational redundancy. If watching-as-relation produces no explanatory or predictive advantage over ordinary causal description, "watcher" is ornamental. §11 is the attempt to meet this. If the ablation dissociation cannot be found in real systems, this fires and takes the paper.

Gate failure. If conditional mutual information is the wrong measure of discriminative gain and no substitute distinguishes recursion that discriminates from recursion that redescribes, the Onion/Watcher distinction collapses. This is a technical claim and can be attacked technically.

Estimator failure. If ΔD cannot be estimated above its own noise floor in the systems where the question actually arises — if every real application lands in the regime where §8's table says the measurement is indistinguishable from zero — the Gate is correct and unusable, which for practical purposes is the same as wrong.

Relativity failure. If the five declarations in §9 cannot be made independently of the desired verdict in practice — if working analysts routinely find that target, ordering, and regime are chosen after seeing the result — then the Gate inherits the arbitrariness §4 was written to prevent, one level up.

Target failure. If §16's boundary is the whole territory — if no case of philosophical interest ever admits a declarable T — the Gate is an engineering criterion in philosophical clothes and the philosophical half

should be withdrawn. The most interesting open threat.

Representation failure. If the misrepresentation criterion admits cases it should exclude or excludes systems that plainly watch, §2 is wrong and the definition falls back to the rock.

Scale arbitrariness. If declaration, bottleneck, and perturbation stability still permit watcher relations to be manufactured by boundary choice, §4 fails.

Cost failure. If observational depth has no identifiable implementation cost in the relevant system, the cost constraint fails there and η is undefined. If the inversion prediction comes out null, cut §6 regardless.

Lineage failure. If the Gate applied to the observer papers in §13 returns a verdict I would not accept from someone else — if I have terminated the layer I was least attached to rather than the layer that discriminates least — then §13 is self-serving and should be rerun by a hostile reader. The check is whether *Those Who Watch Back* can be shown to discriminate something Layer 0 does not. If it can, my verdict is wrong.

Phenomenal smuggling. If functional awareness is used anywhere as evidence for phenomenal consciousness without an independent bridge, the framework has crossed its own boundary and the inference must be rejected, including when I am the one crossing.

Prior art. If a criterion doing the Gate's work already exists as a termination condition for observer regress, this is a rediscovery and should be republished as one with the citation attached.

Closing

The Watcher is not behind the system. It is not above the system. It is not necessarily a thing.

Watching happens when information available from one bounded position becomes causally consequential somewhere else, in a form that survives the source and can be false. Which process counts as watcher depends on scale, boundary, history, and relation — but not on any boundary you care to draw, which is the difference between a relational account and a permissive one.

I spent four years building observers on top of observers before writing down what an observer has to buy. The rule I ended with is the one my own stack had been applying to everything else, in a currency that finally fits the problem that started it. Applied backward, it keeps the layer that added a constraint and terminates the layer that added a mirror.

It is a narrower instrument than it looked when I first wrote it down. It needs a target, an ordering, a regime, a boundary, and enough data to clear its own noise, and it returns nothing trustworthy without all five. That narrowing is not a retreat from the claim. It is the claim — a stopping rule that could not be misapplied would not be measuring anything.

Who watches the watcher becomes three finite questions: from where, at what scale, and what does the watching change. Only the third has a unit. If the answer is under ε, add no watcher.

Gate.

Appendix A — The Calculation

The numbers in §11. Design A is omitted from the code because its value is structural: W_2 is a function of W_1, so T is conditionally independent of W_2 given W_1 and the conditional mutual information is identically zero.

```
import math

# Fault prior
pT = {1: 0.10, 0: 0.90}

# Primary detector: sensitivity 0.80, false-alarm rate 0.10
pW1_T = {1: {1: 0.80, 0: 0.20},
         0: {1: 0.10, 0: 0.90}}

# Design B drift monitor: conditionally independent of W1 given T
pW2_T = {1: {1: 0.60, 0: 0.40},
         0: {1: 0.05, 0: 0.95}}

joint = {(t, w1, w2): pT[t] * pW1_T[t][w1] * pW2_T[t][w2]
         for t in (0, 1) for w1 in (0, 1) for w2 in (0, 1)}

def marginal(idx):
    d = {}
    for k, v in joint.items():
        kk = tuple(k[i] for i in idx)
        d[kk] = d.get(kk, 0.0) + v
    return d

pW1, pW1W2, pTW1 = marginal([1]), marginal([1, 2]), marginal([0, 1])

# I(T ; W2 | W1)
cmi = sum(p * math.log2((p * pW1[(w1,)]) / (pTW1[(t, w1)] * pW1W2[(w1, w2)]))
          for (t, w1, w2), p in joint.items() if p > 0)

# I(T ; W1)
mi = sum(p * math.log2(p / (pT[t] * pW1[(w1,)]))
         for (t, w1), p in pTW1.items() if p > 0)

H = -sum(p * math.log2(p) for p in pT.values())

print(f"H(T)         = {H:.4f} bits")
print(f"I(T;W1)      = {mi:.4f} bits")
print(f"H(T|W1)      = {H - mi:.4f} bits")
print(f"I(T;W2|W1)   = {cmi:.4f} bits   (Design B)")
print(f"residual     = {100 * cmi / (H - mi):.1f}% of what W1 left")
```

Output:

```
H(T) = 0.4690 bits

I(T;W1) = 0.1634 bits

H(T|W1) = 0.3056 bits

I(T;W2|W1) = 0.0791 bits (Design B)

residual = 25.9% of what W1 left
```

The noise-floor, regime, and base-rate tables in §8 and §9 come from a second script, `estimation.py`, which reuses this calculation and sweeps the parameters. Design A's zero is the only value in any of these tables that no parameter choice can move.

Appendix B — Corpus Corrections Arising

Filed here so they do not have to be found.

The Watcher Who Watches the Watchers (April 2025). The limit formulation, and its reading as convergence between observation and existence, is withdrawn — §1. The simulation figure and its caption are withdrawn: the loop contains no watcher term, no stabilizing gain, and no comparison condition, so it cannot show the with-and-without behavior the caption claims — §13. The structural account of recursive observation is retained and superseded by the present paper.

Those Who Watch Back (2025). Terminated under the Gate — §13. The reciprocity construction adds no discrimination over its predecessor, its entropy relations are unbound, and its simulation does not execute as written. One question survives and should be carried forward on its own: whether reciprocal coupling changes what an observer stack can discriminate.

Critique as Architecture (2025). Retained. Its bounded outer loop is the Gate's first form, and §12 identifies it as such. The Lyapunov stopping rule remains correct within its currency; the present paper does not replace it but adds a criterion for the case where a loop converges and buys nothing.

Correction internal to this paper. An earlier draft of §7 listed incremental AUC and system-level ablation as surrogates for ΔD. Both are withdrawn in §8. The second was not merely imprecise: system ablation measures causal contribution, which §7 explicitly distinguishes from discriminative gain and which §11 predicts is non-zero for exactly the layers the Gate terminates. Using it as an estimator would have scored the Onion as valuable. The error is recorded rather than silently repaired because it is the same error the paper is about — a measurement substituted for the thing it was supposed to test.

Prior-art search, 17 August 2026. A deep-research pass was run against the Gate as stated in §7 — a quantitative termination condition for observer regress, with discriminative gain measured as conditional mutual information. It surfaced no prior art. What it returned was the historic neighborhood already named in the body: screening-off, sufficiency, minimum description length.

That is a record of a search, not a finding about the literature. A clean pass means the obvious queries came back empty; it does not mean nothing exists, and it is weakest exactly where the risk is highest — a result stated in different vocabulary, in a methods section, under a title nobody would think to search. I make no novelty claim on the strength of it. The prior-art kill condition in §20 stands unmodified, and anyone who wants to run the search properly is welcome to; if the thing turns up, the condition fires and this paper is a rediscovery.

Sources and Versions

Own corpus. *The Watcher Who Watches the Watchers* (April 2025) and *Those Who Watch Back* (2025) — the observer line this paper supersedes; corrections in Appendix B. *Critique as Architecture* (2025) — the Gate's first form, §12. *The Survival Environment: A Field Binder for the McPhetridge Stack* — Straight-Line Boundary, SCOPE, RRU-C, the drift rule; the Gate's second form. *The Instance Is Not the Model* — trajectory individuation, §14 and §18. *Continuity Without Memory* — the declared-falsifier practice this paper follows.

External. Named in the body by author and claim where they mark a boundary or a neighbor. Each needs locating and checking before publication; none is load-bearing for any argument here.